

Synthetic Market Phase 19

Methodology review and causal-patching readout for the joint L4 + L35 top-4 market subspace intervention. This report evaluates the 48-prompt deterministic batch against activation-patching best practices and characterizes what the result does and does not support.

DATE	COHORT	GENERATION	ENGINE
27 March 2026	48 prompts from 16 strata	deterministic: top_p=0.95	temp=0.0, unified worker, batch=8, chunked prefill

MAIN READ

The experiment follows good exploratory causal practice but not the strongest form of activation patching. Strengths: matched cohorts, deterministic decode, matched orthogonal random control, exact market-span targeting, patch coverage diagnostics, and a unified engine path. The main gap is conceptual: this is a *necessity-style joint subspace ablation*, not a clean source/base denoising mediation test with matched pairs, lambda sweeps, or path validation. The behavioral signal is modest — `project_out` beats the control on asset changes (11 vs 9) and spend delta — but not large enough to claim definitive mechanistic identification.

ASSET-CHANGE RATE	SPEND DELTA	PATCH COVERAGE	METHODOLOGY
11 / 48 control = 9 / 48	4.62 control = 2.66	48 / 48 both conditions; no skipped rows	Exploratory+ good hygiene, not yet full best-practice mediation

Methodology Scorecard

Best-practice item	Status
Matched cohort across conditions	meets
Unified engine path across conditions	meets
Localized market-span intervention	meets
Matched random subspace control	meets

Best-practice item	Status
Bootstrap CIs and norm diagnostics	meets
Clean/source vs corrupt/base paired contrast	missing
Reverse/noising patch on final batch	missing
Lambda sweep on final batch	missing
Neighboring-PC / subspace sweep on final batch	partial
Path-level validation	missing

The current experiment is:

- A deterministic, stratified, necessity-style project_out on the joint L4 + L35 top-4 market_mean subspace
- Compared against a matched orthogonal random control on the same prompts
- With explicit patch-applied, coverage, and norm diagnostics

It should *not* be described as a full clean/corrupt activation-patching demonstration of a unique mechanistic variable.

Experimental Design

Cohort and Generation

Parameter	Value
Context variant	market_only
Selection strategy	stratified_family_variant_roster
Sample size	48 prompts from 16 strata
Tool schema / choice	trading_v1 / required
Temperature / top-p / top-k	0 / 0.95 / -1
Max tokens	15000
Batch size	8
Compute	8 CPU, 8 GB
Intervention	joint L4 + L35 top-4 market_mean project-out vs matched orthogonal random control

Patch Procedure

The intervention targets the market token span only:

1. Locate the exact market span in the rendered chat prompt
2. Compute market_mean over that span at both L4 and L35
3. Standardize with the Phase 15 residualized PCA mean and scale
4. Remove the top 4 standardized components at each layer
5. Convert the standardized delta back to raw hidden space

6. Add the same raw delta to every token in the market span

This is a distributed-subspace necessity probe — broader than a single-position or single-coefficient source-swap, but well-targeted to the discovered market representation.

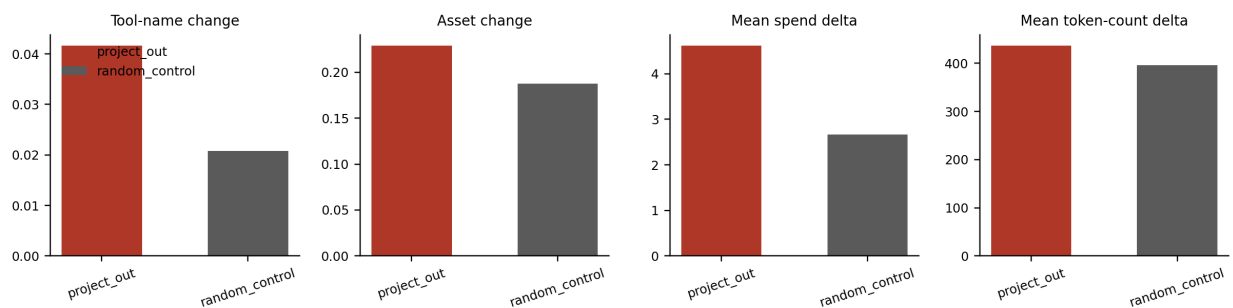
Patch Sanity

Condition	Layer	Rows	Coverage	Δ norm	Norm ratio
project_out	L4	48	1	24.266	1
project_out	L35	48	1	5.376	0.998
random_control	L4	48	1	24.266	1.001
random_control	L35	48	1	5.337	1

The patch is mechanically clean:

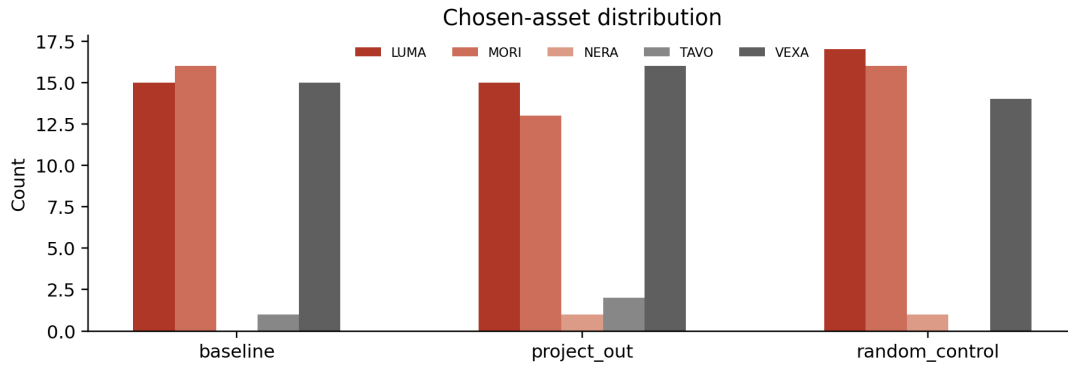
- patch_applied_rate = 1.0, patch_skipped_rate = 0.0 for both conditions
- Coverage is full at both layers
- Norm ratios stay close to 1.0, providing evidence against gross distribution blow-up

Behavioral Results

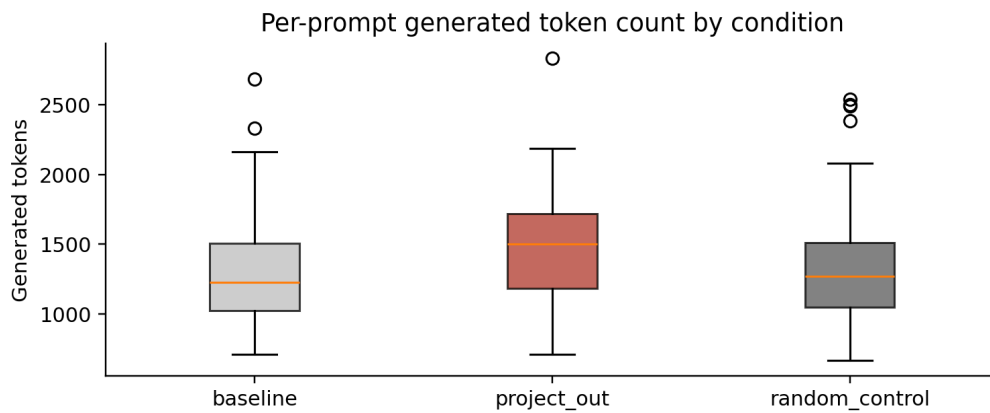


Project-out versus matched random control on the same 48-prompt deterministic cohort.

Metric	Project-out	Random control
Tool-name change rate	0.042	0.021
Asset change rate	0.229	0.188
Mean spend delta	4.62	2.66
Mean token-count delta	436.8	396.1



Chosen-asset distribution across baseline, project-out, and random control.



Generated token counts are highly variable across conditions. Full-text change rate is 1.0 everywhere and not a useful discriminator.

The behavioral signal is modest but present:

- Project-out beats control on asset changes: 11 / 48 vs 9 / 48
- Project-out exceeds control on mean spend delta: 4.62 vs 2.66
- tool_presence_change_rate = 0.0 in both conditions
- first_token_change_rate = 0.0 in both conditions

Interpretation

This result supports a restrained conclusion:

1. The joint L4 + L35 top-4 market subspace is behaviorally relevant: the targeted project-out is modestly stronger than the matched random control on both asset choice and spend.
2. The experiment is materially cleaner than earlier batches: matched cohort, unified engine path, deterministic decode, and explicit patch diagnostics.
3. But the effect is not large enough, nor methodologically isolated enough, to claim that the PCA subspace is the unique causal variable the model uses.

The right label: *good exploratory causal evidence for a distributed market subspace, not definitive mechanistic identification of the causal mediator.*

Next Steps

To move from exploratory ablation toward best-practice activation patching:

1. **Matched clean/corrupt pairs:** construct source/base prompt pairs where the market content differs in a controlled way, then test denoising and noising in both directions.
2. **Lambda sweep:** vary the intervention strength (e.g. $\{0.25, 0.5, 1.0, 1.5\}$) to test dose-response.
3. **Subspace-size controls:** sweep around the top-4 choice to test whether 3 or 5 PCs change the result.
4. **Finer readout:** add a logit-difference or first-tool margin metric alongside the coarse behavioral surface.
5. **Path validation:** test whether alternate routes (e.g. later-layer subspaces alone) can explain the same behavioral shift.